

TITLE: - LPC DECODE IO RANGE

AIM: -

To List LPC decode IO range by Assembly program

ALGORITHM: -

- Initialize data segment.
- Display start message.
- Load PCI address into EAX.
- Write address to port 0CF8h.
- Read data from port 0CFCh into EAX.
- Convert EAX value to hexadecimal string.
- Display result message.
- Display hexadecimal value.

PROGRAM:-

TITLE LPC.ASM -- Read and display LPC Decode IO Range

```
;/******  
;/* *  
;/* Copyright (c) 1985-2022, American Megatrends International LLC. *  
;/* *  
;/* All rights reserved. Subject to AMI licensing agreement. *  
;/* *  
;/******  
;<AMI_FHDR_START>  
;  
; Header:  
; Author: Muthuganesh s  
; Description: Read LPC GEN1_DEC Register (Bus 0, Dev 31, Func 0, Offset 84h)  
; Date: 29/03/2026
```

```
;
;<AMI_FHDR_END>

,*****
,

.model small

.386      ; REQUIRED: Allows use of 32-bit registers (EAX) in 16-bit mode

.stack 100h

;-----
;   DATA SEGMENT
;-----

data segment

    msg_start db 13, 10, "Reading LPC GEN1_DEC Register (Bus 0, Dev 31, Fun 0, Off 84h)...$"
    msg_res   db 13, 10, 13, 10, "GEN1_DEC Value: 0x$"
;-----
; Buffer for 32-bit Hexadecimal Output String
;-----
    hex_buf   db 8 dup('0') ; 8 bytes for hex characters
               db '$'       ; DOS string terminator

data ends

;-----
;-----
;   CODE SEGMENT
;-----

code segment

assume cs:code, ds:data

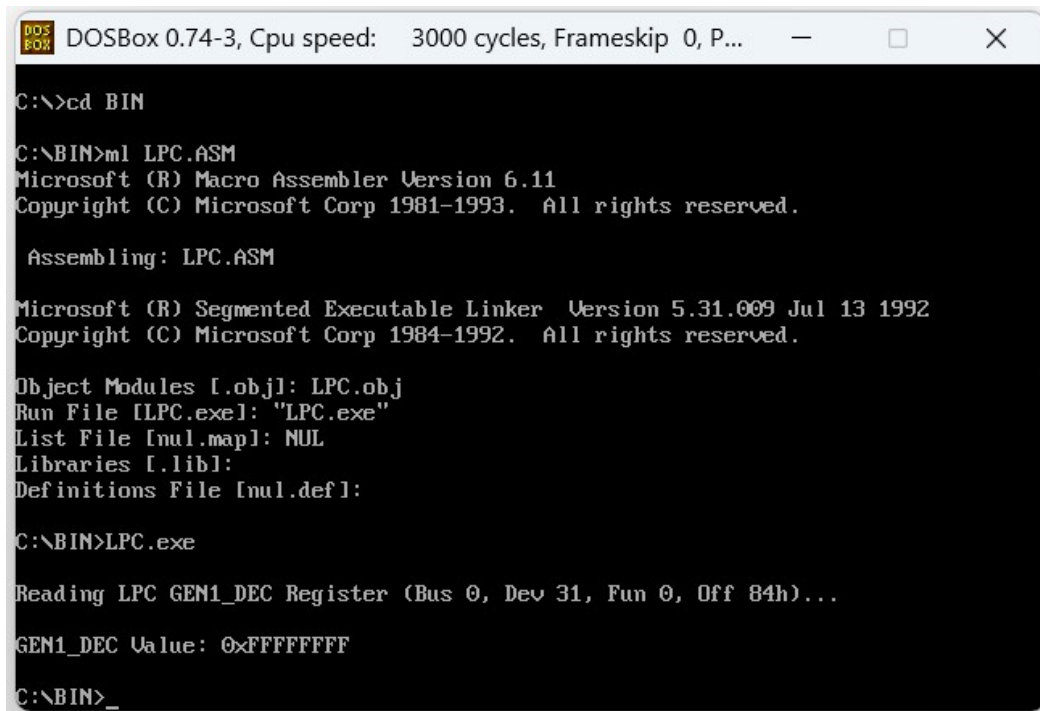
start:
```

```
;-----  
; Initialize Data Segment  
;-----  
  
mov ax, data  
mov ds, ax  
  
;-----  
; Print Start Message  
;-----  
  
mov dx, offset msg_start  
mov ah, 09h  
int 21h  
  
;-----  
; 1. Calculate and Write PCI Configuration Address  
; Format: Enable (Bit 31) | Bus 0 | Dev 31 | Func 0 | Offset 84h  
; Enable = 80000000h  
; Dev 31 = 0000F800h  
; Offset = 00000084h  
; Address = 8000F884h  
;-----  
  
mov dx, 0CF8h      ; Point DX to PCI Config Address Port  
mov eax, 8000F884h ; Load the GEN1_DEC address  
out dx, eax        ; Write address to port  
  
;-----  
; 2. Read the PCI Data  
;-----  
  
mov dx, 0CFCh      ; Point DX to PCI Config Data Port
```

```
in eax, dx      ; Read 32-bit register value into EAX
;-----
; 3. Convert EAX (Binary) to Hexadecimal String
;-----
mov di, offset hex_buf ; Point DI to our output buffer
mov cx, 8          ; 8 nibbles (hex digits) in a 32-bit register
hex_convert_loop:
    rol eax, 4      ; Rotate highest 4 bits (nibble) into AL
    push eax        ; Save the full EAX state
    and al, 0Fh     ; Mask out everything except the bottom nibble
    add al, '0'     ; Add ASCII '0'
    cmp al, '9'     ; Is it a number (0-9)?
    jle store_hex_char ; If yes, jump to store it
    add al, 7       ; If no, add 7 to convert to 'A'-'F'
store_hex_char:
    mov [di], al    ; Write the ASCII character to the buffer
    inc di          ; Move to the next byte in the buffer
    pop eax         ; Restore EAX for the next rotation
    loop hex_convert_loop ; Decrement CX, loop if not zero
;-----
; 4. Print Result Message and Hex Value
;-----
mov dx, offset msg_res
mov ah, 09h
int 21h
mov dx, offset hex_buf
mov ah, 09h
int 21h
; Print a couple of newlines for clean exit
```

```
mov dl, 13
mov ah, 02h
int 21h
mov dl, 10
mov ah, 02h
int 21h
;-----
; Exit program
;-----
exit_program:
    mov ah, 4Ch
    int 21h
code ends
end start
;-----
```

OUTPUT: -



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, P...
C:\>cd BIN
C:\BIN>ml LPC.ASM
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: LPC.ASM

Microsoft (R) Segmented Executable Linker Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp 1984-1992. All rights reserved.

Object Modules [.obj]: LPC.obj
Run File [LPC.exe]: "LPC.exe"
List File [nul.map]: NUL
Libraries [.lib]:
Definitions File [nul.def]:

C:\BIN>LPC.exe

Reading LPC GEN1_DEC Register (Bus 0, Dev 31, Fun 0, Off 84h)...

GEN1_DEC Value: 0xFFFFFFFF
C:\BIN>_
```